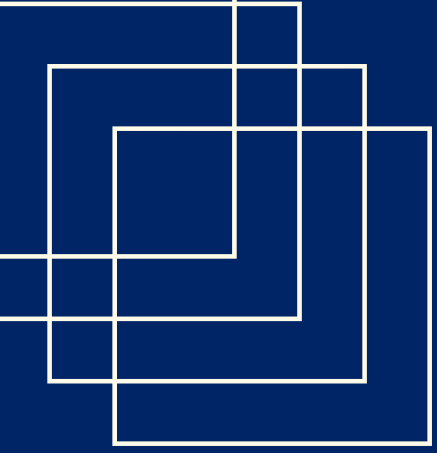


RAG: EARLY CHILDHOOD DEVELOPMENT MILESTONES

**GENAI GROUP PROJECT
SPRING '26
GROUP 7**



EARLY CHILDHOOD DEVELOPMENT MILESTONES CHATBOT

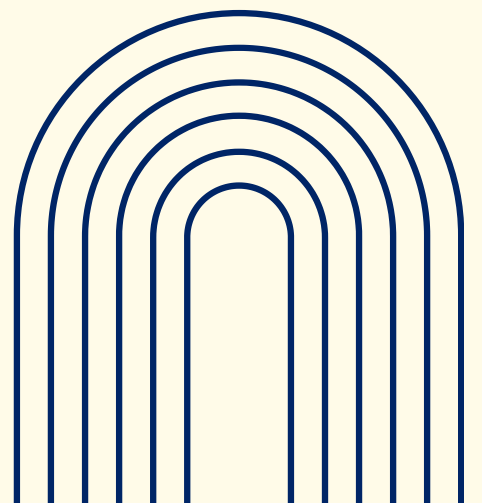
Objective: To build a domain-specific chatbot providing reliable information on early development milestones (2 months to 2 years) using CDC and UNICEF data.

The chatbot is intended to answer questions about general developmental patterns, including language development, social interaction, emotional development, and how children explore and learn.

The goals for our chatbot:

- For informational purposes only.
- It does not provide medical advice, does not diagnose developmental conditions, and does not replace professional medical, psychological, or developmental assessment.

This task is performed using a Retrieval-Augmented Generation (RAG) pipeline supported by a verified knowledge base, ensuring information is evidence-based rather than synthesized from general training data.



TECHNICAL IMPLEMENTATION & PIPELINE ARCHITECTURE

FROM INDEXING TO PERFORMANCE EVALUATION

Knowledge Base

Data was scraped and structured into a JSON format, converted into LangChain Document objects, and stored in a Chroma Vector Database.

Embedding & Retrieval

Used the all-MiniLM-L6-v2 model for semantic search, configured to retrieve the top 5 most relevant context chunks (k=5).

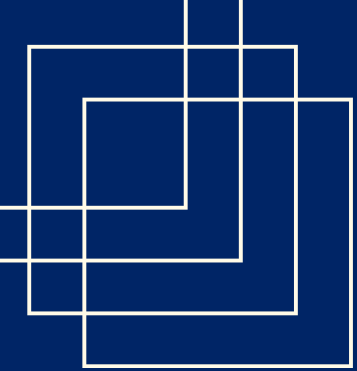
Model Selection

Tested both Qwen-2.5-3B and Microsoft Phi-3-mini. The Phi-3 model was selected for the final build due to its efficiency.

Safety

Pre-screens queries for medical keywords and automatically appends legal disclaimers or fallback responses to prevent unauthorized medical advice.

Once implementation was completed, UI enhancements including chat management features and suggested prompts were added. Finally a RAGAS evaluation framework was executed to understand system performance and potential improvements.



UNDERSTANDING PERFORMANCE USING RAGAS (RETRIEVAL-AUGMENTED GENERATION ASSESSMENT)

The evaluation results indicated that our system is performing well when it its finding information (retriever performance) yet it is struggling to process and deliver the relevant chunks in accordance with the specific query.

- **Context Recall (0.60):** Our Context Recall score was the highest of the four metrics, which tells us that the retriever is successfully finding the correct information in the database for most of the supplied questions.
- **Context Precision (0.39):** Our Context Precision score indicates that while a contextual response for a user's query is being retrieved, the most relevant chunk is not always being ranked accordingly.
- **Faithfulness (0.32):** Our low Faithfulness score implies the judge found many of the chatbot's claims were not actually supported by the retrieved context chunks, suggesting the model may be hallucinating or using outside knowledge despite the guardrails implemented.
- **Answer Relevancy (0.16):** Our low Answer Relevancy score indicates that the final answers often fail to directly address the supplied prompt, showing the Phi-3-mini model is struggling to synthesize the provided context from the retriever.

LEARNINGS & NEXT STEPS

Our analysis using the RAGAS framework indicated limits in smaller language models. While our Chroma database demonstrated a strong ability to locate specific milestones (0.60 Context Recall), the transition from stored data to a generated response remains the primary area for growth. The lower scores in Answer Relevancy (0.16) and Faithfulness (0.32) help us to understand that SLMs like Phi-3-mini require significantly more prompt engineering and noise reduction to perform at the same scale of LLMs.

A highlight of our implementation was the adherence of medical guardrails. While understanding that medical issues (especially those for children) are not to be addressed without a doctor's examination, we've successfully integrated our safety parameters into a tool that could help parents and caregivers learn about their child's development roadmap.

If the chatbot were to be developed for a production use case, some options to enhance the chatbot would include:

- Retriever changes could include integrate keyword-based search with our existing semantic search to ensure a more accurate retrieval for specific developmental terms.
- Generator updates would be moving away from an SLM toward a larger model containing more parameters and higher performing attention mechanisms to enhance the “Faithfulness” and “Answer Relevancy” we currently operate with.

